جامعة محمد بن زايد للذكاء الاصطناعي
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Comparative Analysis of Datashift/Distribution shift Effects on Model Generalization: Hospitality Domain

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Task Introduction: Objective of the Project

Project Overview:

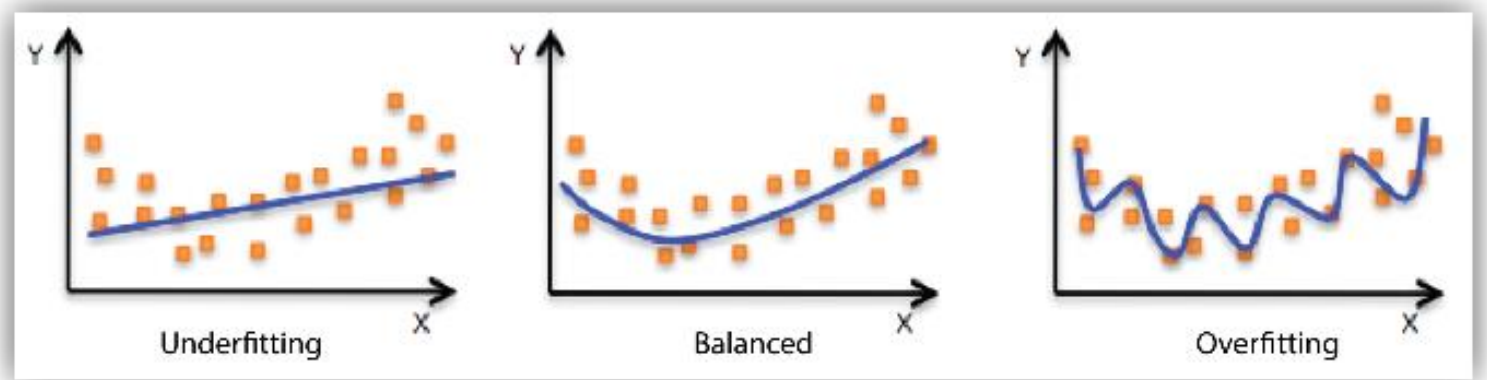
- **Purpose:** Examine the adaptability and performance of various machine learning models in response to data shifts within the hospitality industry.
- **Context:** The hospitality industry faces dynamic changes in market conditions, influenced by factors like economic fluctuations, seasonal trends, and changing customer preferences.

Challenges Addressed:

- **Data Shifts:** Explore how shifts in the statistical properties of data, due to temporal and geographic variations, affect predictive model performance.
- **Model Robustness and Generalization:** Assess models on their ability to deliver reliable outputs across varying conditions and their capability to perform accurately on new, unseen data.

Importance of Study:

- **Operational Efficiency:** Enhancing predictive accuracy helps in optimizing resource allocation, pricing strategies, and customer satisfaction in the hospitality sector.
- **Strategic Decision Making:** Reliable models support more informed decision-making, critical in managing the unpredictability of the hospitality market.

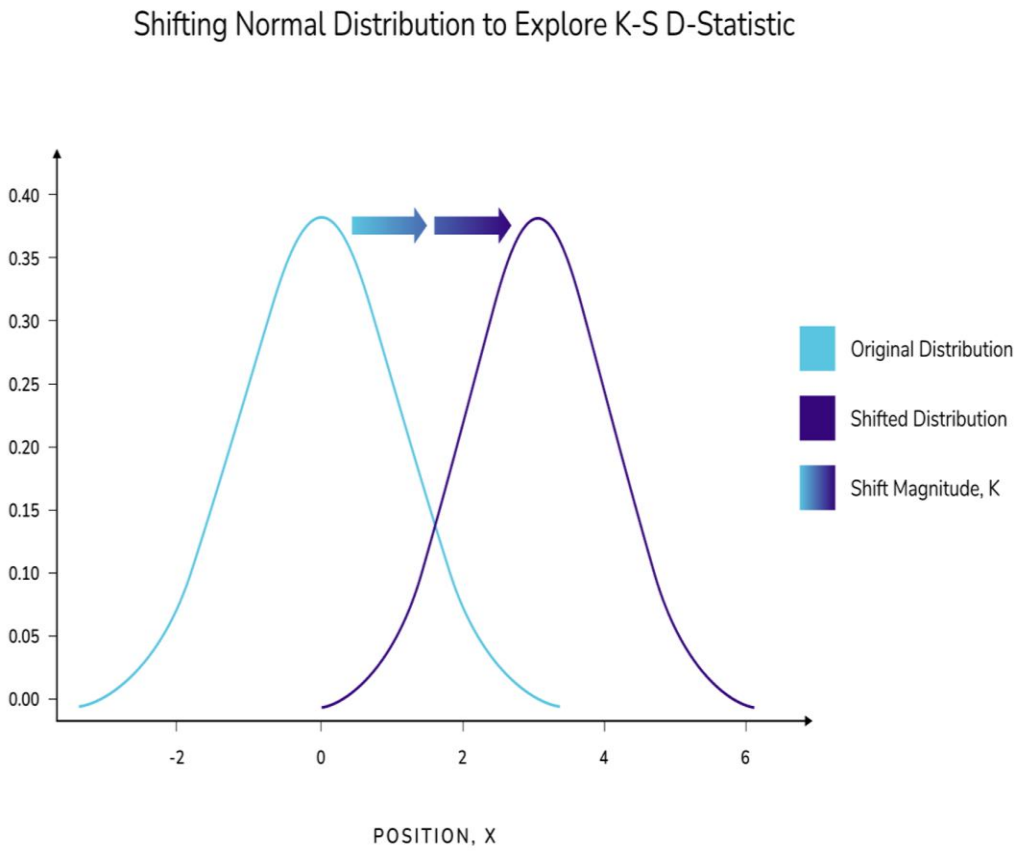


Understanding Data Shifts

- **What is a Data Shift?**
  - ✓ **Definition:** Data shifts occur when the statistical properties of input data change between the training phase of a model and its application phase. This can result in the model performing differently than expected.

- **Types of Data Shifts:**
  - ➢ **Covariate Shift:** When the distribution of input features change
  - ➢ **Label Shift:** When the distribution of output labels changes.
  - ➢ **Concept Drift:** When the relationship between the input featur output labels changes over time.

- **Causes of Data Shifts in Hospitality:**
  - ➢ **Seasonal Variations:** Changes in customer behavior and prefer different times of the year.
  - ➢ **Economic Factors:** Impact of economic fluctuations on custom and travel habits.
  - ➢ **Emerging Trends:** New customer preferences and competitive s altering the dynamics of the market.

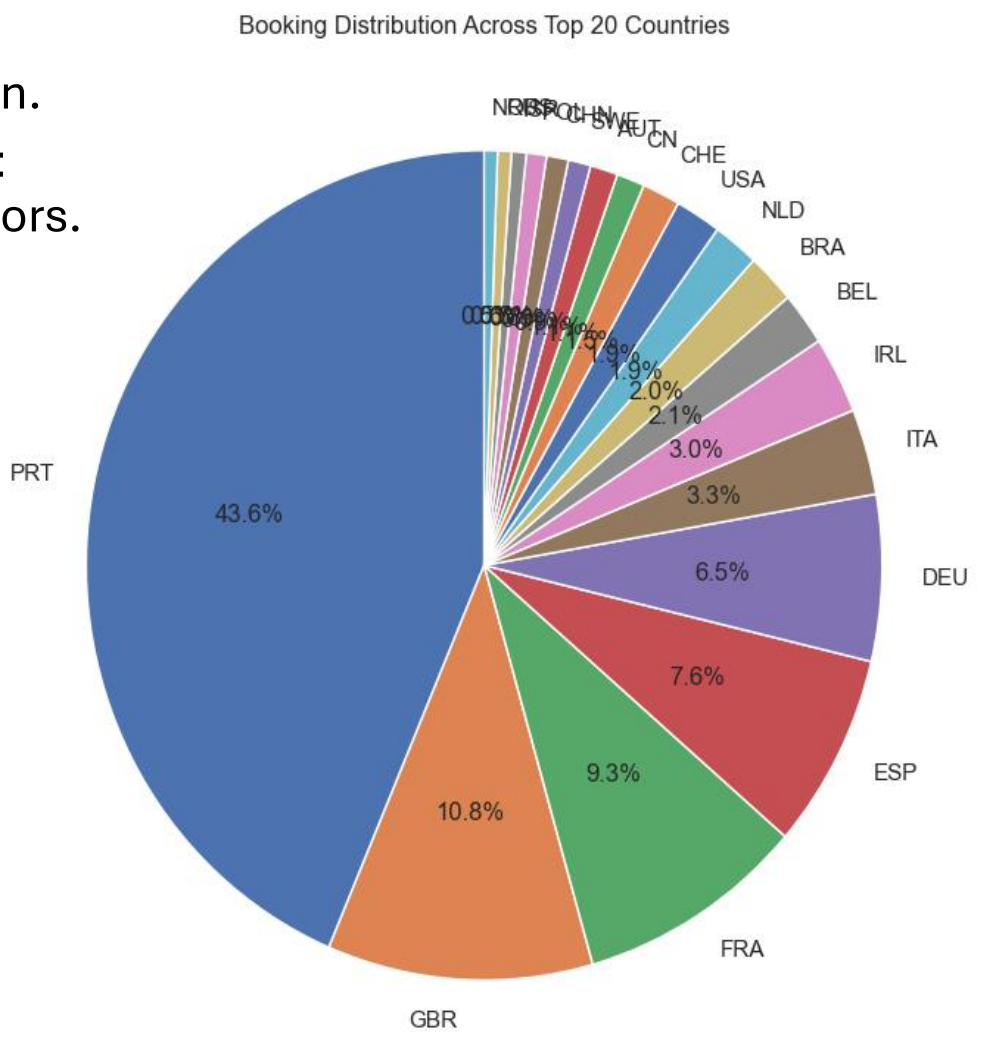


- **Impact on Predictive Models:**
  - ➢ **Performance Degradation**: Models may perform well on historical data but fail to predict accurately on current data if not adjusted for shifts.

Methodology Overview

1. Dataset Description and Preparation

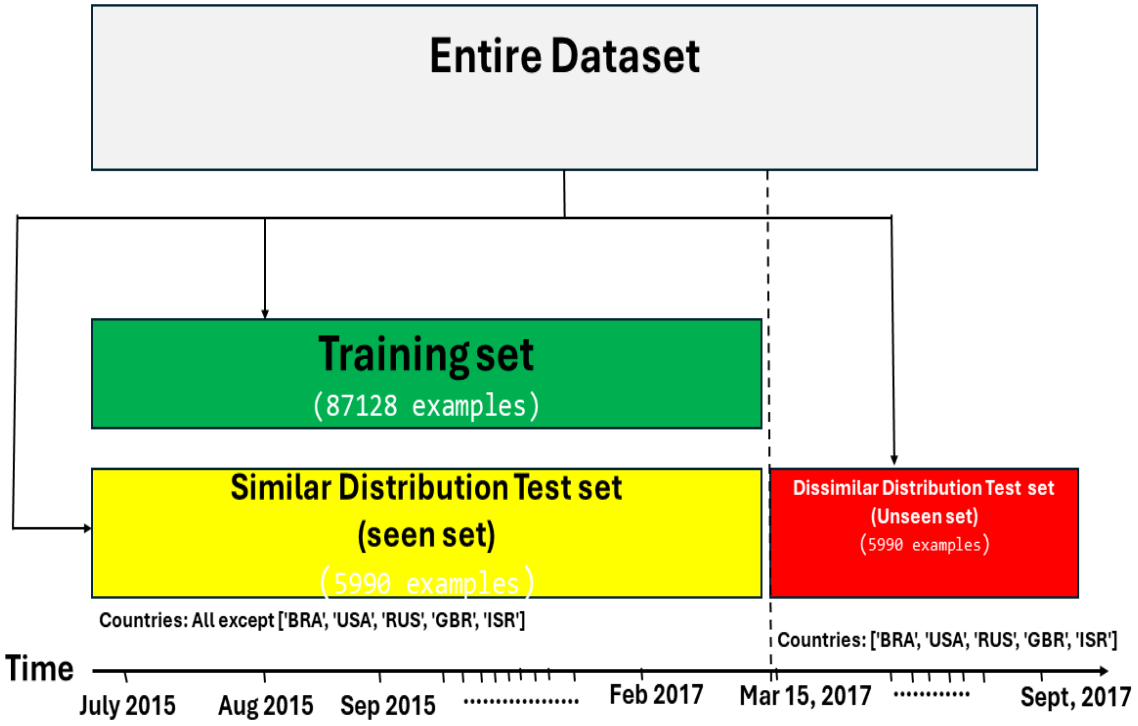
- **Dataset Overview:**
  - **Source:** Over 119,000 booking records from a global hotel chain.
  - **Coverage:** Data includes information from **July 2015 to August 2017**, covering diverse market conditions and customer behaviors.
  - **Location:** Collected from 20 different countries
- **Data Features:**
  - **Target variable**: Predict if a customer will cancel their booking
  - **Key Variables:** Lead time, duration of stay, number of guests, booking cancellations, and economic indicators.
  - **Relevance:** Each feature selected based on its influence on booking dynamics and potential to reflect shifts in customer preferences and market conditions.
- **Data Cleaning and Normalization:**
  - Handling missing values, removing outliers, and standardizing data to ensure uniformity across different datasets.



Methodology Overview

2. Data Splitting Strategy

- **Purpose of Data Splitting:**
  - **Objective:** Create realistic training and testing scenarios that simulate both familiar and unfamiliar market conditions.
- **Data Splitting Approach:**
  - ➢ **Training Set**: Includes data up to March 15, 2017, excluding data from regions known for high variability to create a stable training environment.
  - ➢ **Seen Test Set:** Comprises data post-March 15, 2017, that follows the same distribution as the training set to evaluate model performance under known conditions.
  - ➢ **Unseen Test Set:** Consists of data from previously excluded regions, introduced post-March 15, 2017, to test model robustness against new and unexpected data shifts.



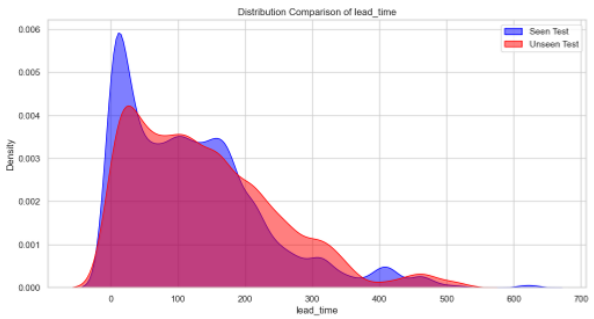
Methodology Overview

3. Quantifying Data shifts as a Result of Splitting Strategy

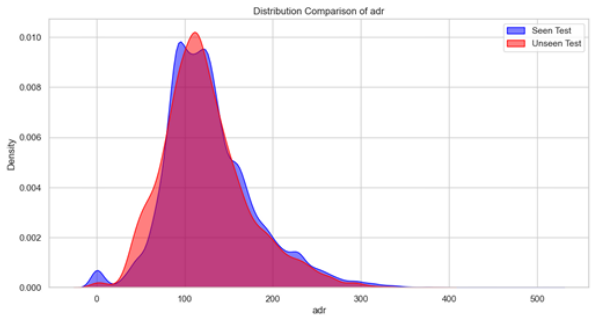
- **Purpose of Quantification:**
  - **Objective:** Ensure there are actual distribution shifts between the seen and unseen test sets to validate the testing methodology.
- **How?**
  - **Kolmogorov-Smirnov Test:** Applied to detect and quantify distribution shifts across key variables such as lead time, average daily rate (ADR), total guests, and total stay.
- **Main contributors to data shift**
  - Variables Analyzed:
  - Lead Time ***(Most correlated variable)***
  - Average Daily Rate (ADR)
  - Total Guests
  - Total Stay

Table 1: Summary of KS Test Results

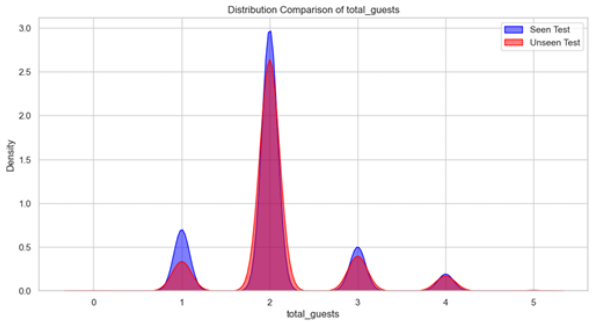
Variable	KS Statistic	P-value
Lead Time	0.0803	1.886×10^{-27}
ADR	0.0515	1.533×10^{-11}
Total Guests	0.0652	3.054×10^{-18}
Total Stay	0.0910	3.976×10^{-35}



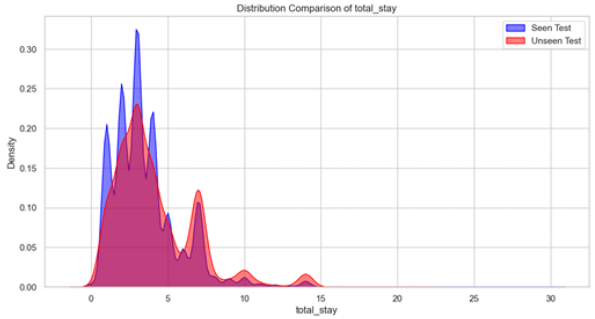
(a) Lead time distribution shift



(b) ADR Distribution shift



(c) Total number of guest distribution shift



(d) Total stay distribution shift

Methodology Overview

4. Model Selection and Rationale

- **Overview of Selected Models:**
  - **Range of Complexity:** From simple logistic regression to complex convolutional neural networks (CNNs), encompassing a broad spectrum of machine learning methodologies.
- **Criteria for Model Selection:**
  - **Adaptability to Data Shifts:** Models were chosen based on their potential to handle changes in data distributions effectively.
  - **Robustness and Accuracy:** Ability to maintain performance consistency across different data scenarios.
  - **Computational Efficiency:** Consideration of the models' demands on computational resources relative to their performance benefits.
- **Expected Contributions of Each Model:**
  - **Assess Generalization Capabilities:** How well each model performs under the conditions of seen and unseen data sets.
  - **Understanding Model Sensitivity:** Identify which models are most sensitive to shifts in input data and which are robust against such variations.

Table 2: Summary of Models Used in the Study

Model	Total Parameters
Logistic Regression	249
Random Forest	10,000
Gradient Boosting	28,000
Bayesian Neural Network	40,193
DNN	291,841
CNN	2,011,713

Experimental Setup and Evaluation Metrics

- **Training and Testing Protocol:**
  - **Training Data:** Utilized records up to March 15, 2017. Excluding countries [’BRA’, ’USA’, ’RUS’, ’GBR’, ’ISR’].
  - **Same Distribution Test Set:** Data up to March 15, 2017, matching training distribution. Excluding countries [’BRA’, ’USA’, ’RUS’, ’GBR’, ’ISR’].
  - **Shifted Distribution Test Set:** Data from excluded regions, testing robustness.
- **Key Evaluation Metrics:**
  - **Accuracy:** Correct predictions/Total observations.
  - **AUC-ROC**: Trade-off between true positive rate and false positive rate.
  - **Brier Score:** Mean squared difference between predicted probabilities and actual outcomes. *Brier Score closer to zero implies the closer the predicted probabilities to the actual outcome.*
  - **Negative Log-Likelihood (NLL):** Measures model's predictive accuracy for probabilistic outputs.
- **Performance Analysis:**
  - **Comparative Metrics:** Analyze performance differences on seen vs. unseen datasets.

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2$$

forecasted probability

actual outcome of the event

Experiment Results

Overview of Model Performance:

- **Logistic Regression:** Served as the baseline for performance comparison.
- **Random Forest and Gradient Boosting:** Exhibited robust performance on seen data; variations noted on unseen data.
- Despite more complex than RF and GB, Bayesian Neural Network, yields poor results unlike expected, **but WHY??**
- **CNN:** Demonstrated the highest resilience and best overall performance, especially on unseen datasets.

Table 3: Experiment Results: Accuracy and ROC

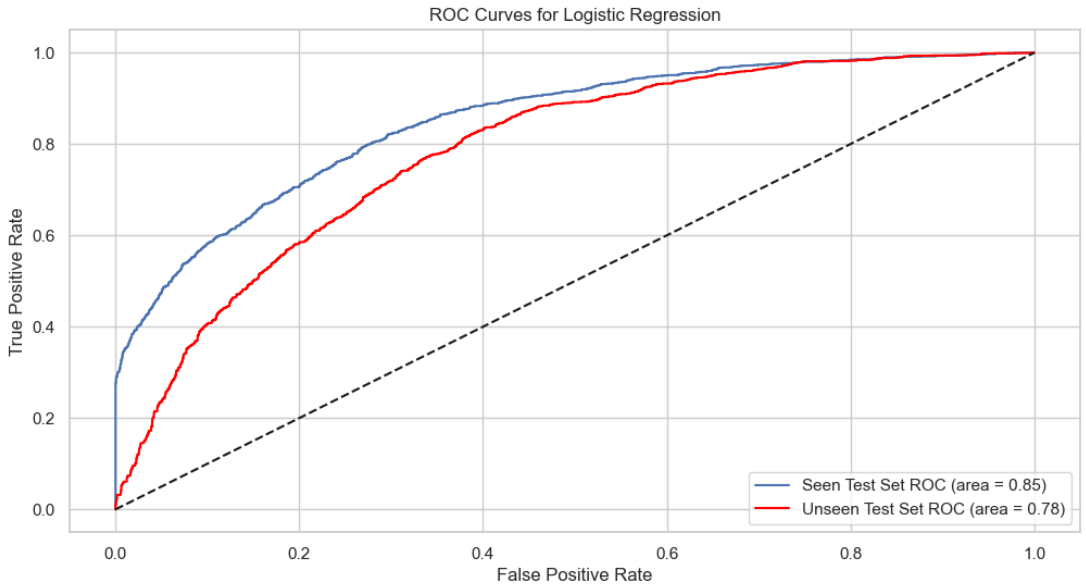
Model	Total Parameters	Accuracy Seen (%)	Accuracy Unseen (%)	ROC Difference
Logistic Regression	249	76.89	72.46	0.07
Random Forest	10,000	83.38	79.01	0.02
Gradient Boosting	28,000	84.29	80.79	0.03
Bayesian Neural Network	40,193	83.49	78.72	0.09
DNN	291,841	87.53	83.86	0.06
CNN	2,011,713	90.18	87.15	0.05

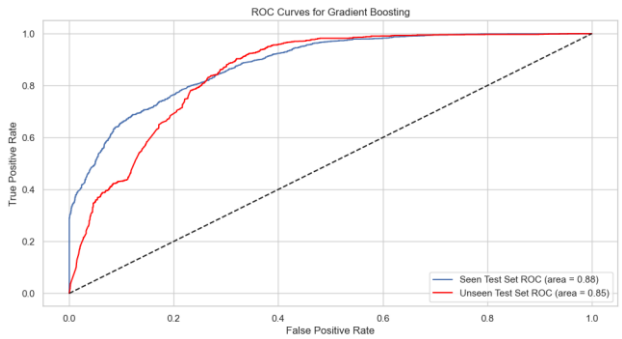
Table 4: Experiment Results: Brier Score, NLL, and Differences

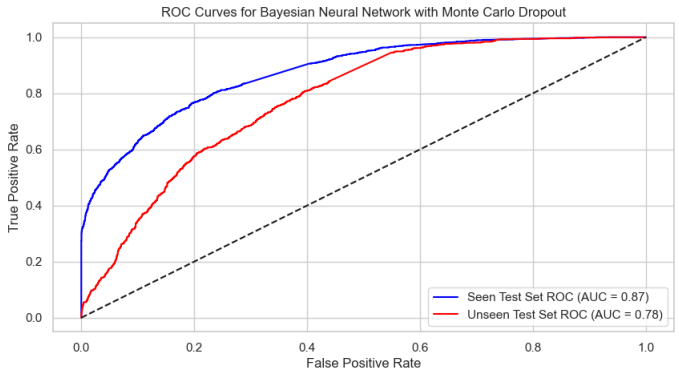
Model	Total Parameters	Brier Score (Seen, Unseen)	Brier Difference	NLL (Seen, Unseen)	NLL Difference
Logistic Regression	249	0.1528, 0.1677	0.0149	0.4594, 0.5022	0.0428
Random Forest	10,000	0.1404, 0.1577	0.0173	0.4224, 0.4640	0.0416
Gradient Boosting	28,000	0.1384, 0.1521	0.0137	0.4167, 0.4530	0.0363
Bayesian Neural Network	40,193	0.1496, 0.1817	0.0321	0.4507, 0.5387	0.0880
DNN	291,841	0.1570, 0.1676	0.0106	0.4730, 0.5082	0.0352
CNN	2,011,713	0.1419, 0.1461	0.0042	0.4309, 0.4457	0.0148

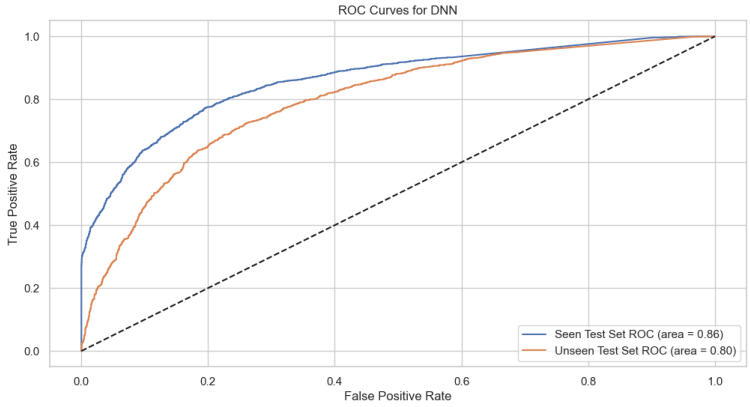
Experiment Results

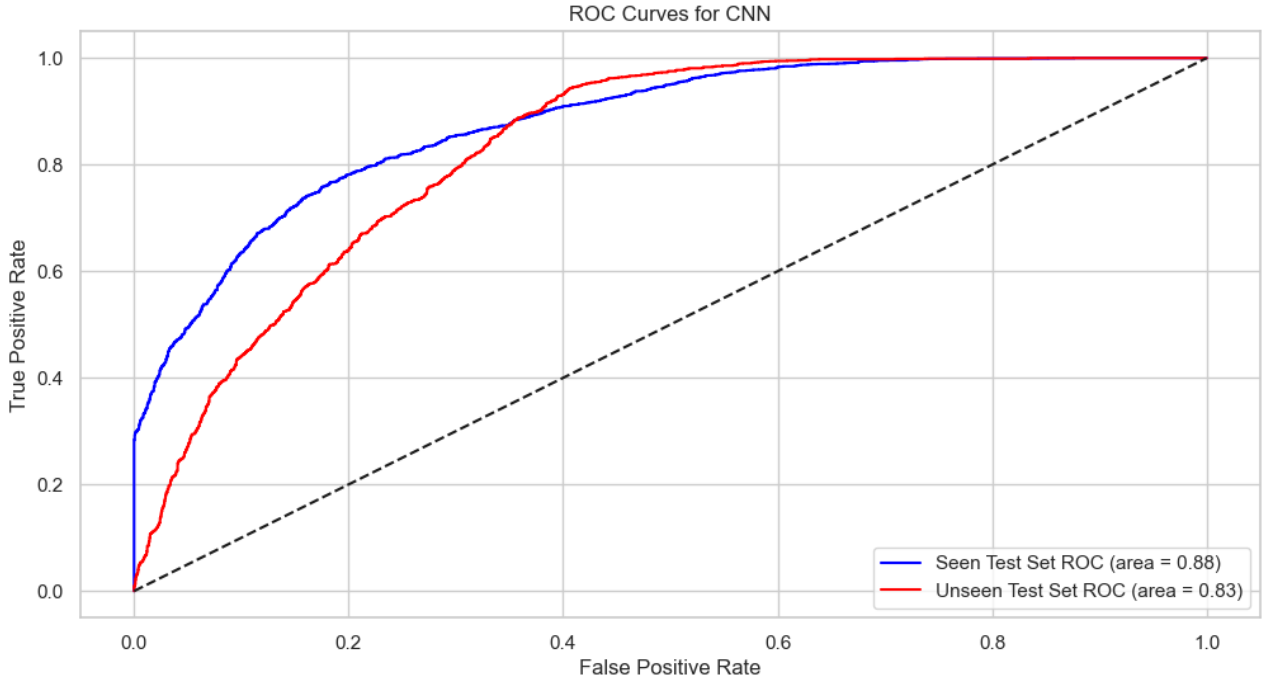
Overview of Model Performance: ROC Plot











Discussion of Results: Double Descent Phenomenon

- **Double Descent Overview:**
  - **Concept:** Model performance dips and then improves as complexity increases from overfitting to recovery.
- **Model Performance Analysis:**
  - **Initial Descent:** Seen in simpler models like Logistic Regression and Random Forest.
  - **Peak of Double Descent:** Highlighted by performance drops in the Bayesian Neural Network.
  - **Second Descent:** Performance improves significantly in models like DNN and CNN.
- **Graphical Illustrations (Next Page):**
  - **Graph 1:** Display differences in AUC, Brier, and NLL between seen and unseen data.
  - **Graph 2:** Show the 'Double Descent: Accuracy vs Model Complexity' curve, indicating key performance changes.

Discussion of Results: Double Descent Phenomenon

- **Graphical Illustrations:**
  - **Graph 1:** Display differences in AUC, Brier, and NLL between seen and unseen data.
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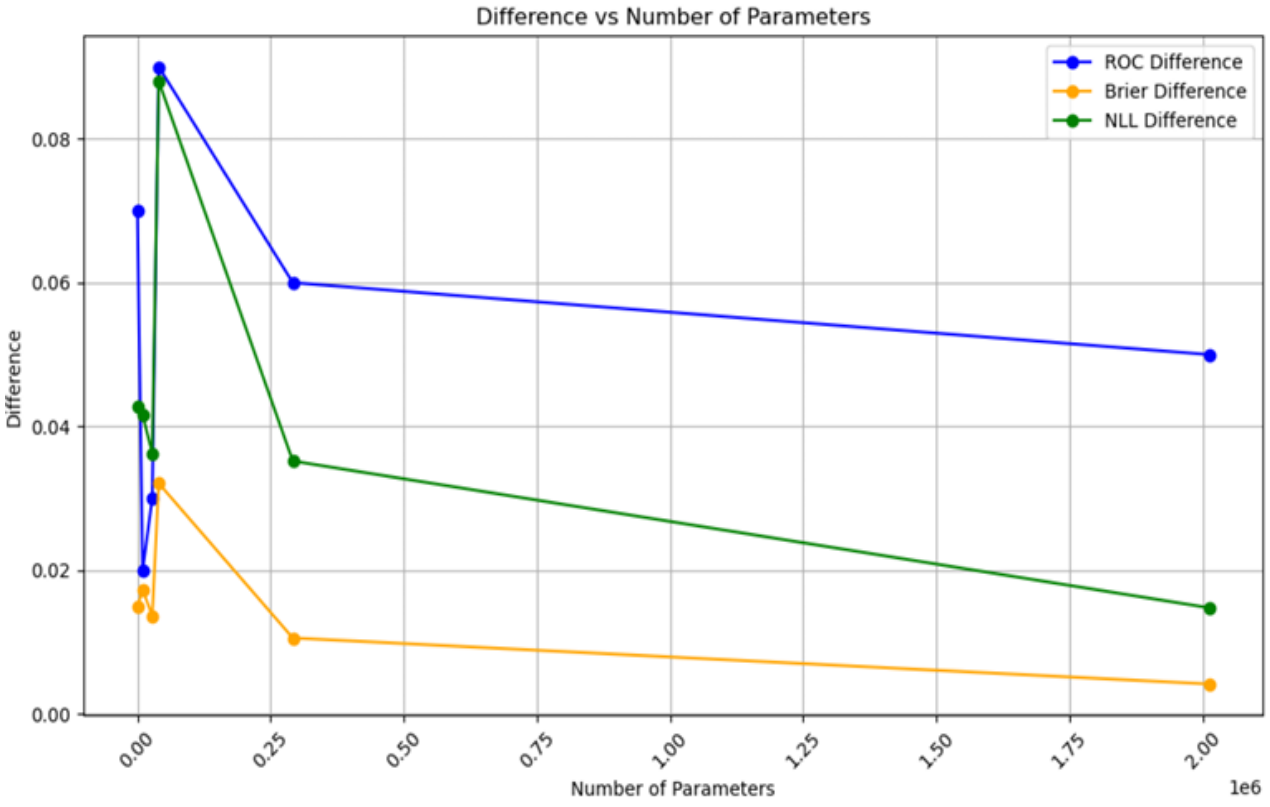

Figure 7: AUC, Brier, and NLL difference between seen and unseen for each model

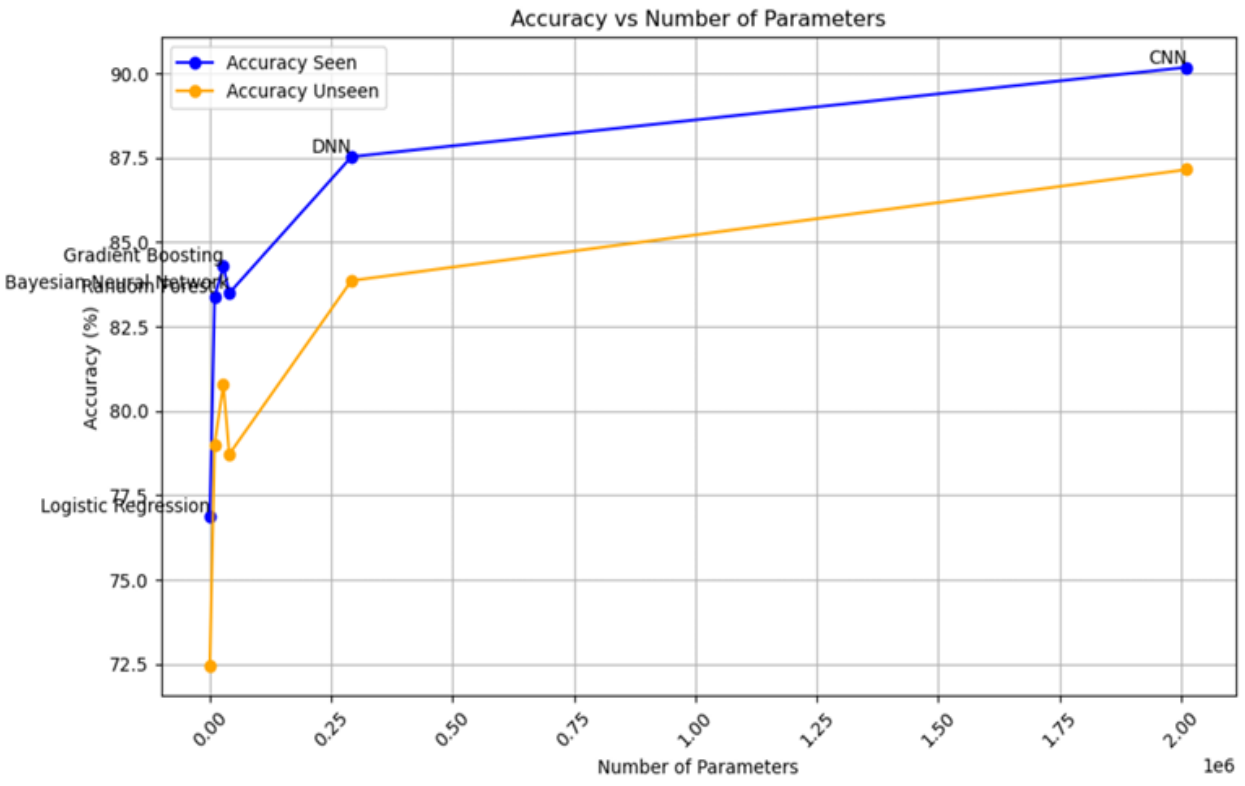

Figure 8: Double Decent: Accuracy vs Model complexity

Conclusion

- **Key Takeaways:**
  1. **Understanding of Data Shifts:** Demonstrated the significant impact of data shifts on model performance, emphasizing the need for models that adapt to changing conditions.
  2. **Model Performance Across Data Shifts:** Highlighted how complex models, particularly CNNs, are more adept at handling unseen data, providing robustness that is crucial for dynamic market conditions.
  3. **Double Descent Phenomenon:** Explored the implications of model complexity on performance, with a focus on the challenges and benefits at different stages of complexity.
- **Contributions to the Field:**
  1. **Advancing Model Selection:** Provided insights that aid in the selection and deployment of machine learning models in industries characterized by frequent data shifts.
  2. **Enhancing Predictive Accuracy:** The findings contribute to the development of more accurate and reliable predictive tools, which are vital for strategic decision-making in the hospitality sector.
- **Final Thoughts:**
  1. **Continued Research Importance:** Stress the importance of ongoing research to further understand and mitigate the effects of data shifts on predictive modeling.
  2. **Practical Implementation:** Encourage the integration of these findings into practical applications, ensuring that models remain effective as data environments evolve.

Thank you! Questions??

References

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